Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/20/2025
PI Name:	Cornforth, Michael Ph.D.		
Project Title:	Molecular Characterization of Transmissible Chromosome Aberrations Produced by Ions of Intermediate and High Atomic Number: grant 80NSSC21K0679		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/11/2021	**End Date:**	03/10/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/10/2025 per V. Lehman/ NASA-JSC Grants Office (Ed., 4/16/24).		
Key Personnel Changes/Previous PI:	Update per the Principal Investigator (PI): Yongjia Yu, Ph.D. is no longer with the project as CoInvestigator (Ed, 3/22/23).		
COI Name (Institution):			
Grant/Contract No.:	80NSSC21K0679		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation as directed research of "Molecular Characterization of Transmissible Chromosome Aberrations Produced by Ions of Intermediate and High Atomic Number," grant NNX15AG74G; Principal Investigator Dr. Michael Cornforth continues as PI. Risks associated with crewed space flight involve extremely low fluences of mixed-field galactic cosmic radiation (GCR), and secondary neutrons formed through the interaction of lighter ions with the structure of spacecraft. It is simply not possible to duplicate this environment in any biologically meaningful experimental setting on Earth. The best solution to this problem is to simulate key aspects of the deep space environment by focusing on “representative” aspects of GCR. For obvious reasons, it is not feasible to measure directly biological risk to astronauts. Thus, we are forced to employ experimental model systems that mimic central aspects of the deep space radiation environment, and which employ surrogate biological endpoints of principle relevancy to biological endpoints of concern, like cancer. Most all ionizing radiation (IR)-induced chromosomal aberrations are exchanges that occur between different chromosome (e.g., translocations), or between regions within the same chromosome (e.g., inversions). Beyond the basic investigative platform that it provides to the science of radiation biology, knowledge concerning the genesis of IR-induced breakpoint junctions carries specific potential benefits. As for NASA and its priorities are concerned, a more detailed view of the enzymology of the rejoining reactions themselves would be of immense value to modelers, particularly as it relates to ultra-low dose risk assessment. Another benefit lies in the prospect that changes in radiation quality will influence the molecular structure of IR-induced breakpoint junctions. Colonies derived from the single-cell expansion of irradiated human telomerase reverse transcriptase (hTERT)-immortalized BJ-1 fibroblast cultures form the basis of this system. These cytogenetically characterized clones form the starting material for subsequent sequencing analysis. Specific Aims Aim 1: Determine if analysis of IR-induced chromosome aberration (CA) breakpoint junctions suggest different molecular pathways are involved, with respect to high- versus low Linear Energy Transfer (LET) exposure. Aim 2: Determine if the sizes of co-deleted sequences at breakpoint junctions differ for those produced by high versus low LET IR; these may form the basis of an LET-dependent biomarker. Aim 3: Determine if the breakpoints associated with complex chromosomal exchanges are different than those associated with simple reciprocal exchanges that involve only two chromosomes. Aim 4: Examine the breakpoint junctions of exchange aberrations induced by medium-to-high energy neutrons. Aim 5: Characterize molecular differences between the breakpoints produced by different ultra-low dose ions.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This study represents a continuation of Chromosome Aberrations I (NNX15AG74G) project; the objectives and research goals remain largely unchanged. We seek fundamental information at the molecular level about the nature of chromosome aberration breakpoints induced by ionizing radiations (IR) that differ vastly in ionization density, comparing terrestrial gamma rays to those of much higher linear energy transfer rates. To accomplish this will require further development and refinement of an experimental model system for which we have previously established proof-of-principle efficacy. This includes implementation of advanced methods of massively parallel sequencing such as single molecule/real-time sequencing.
Research Impact/Earth Benefits:	Radiation-induced reciprocal chromosome translocations and inversions are particularly important in that regard as they relate to crewed space activities. In addition to causing cancer, their appearance also accompanies ongoing genome instability processes associated with their progression. The fact that these particular chromosome aberrations are transmissible (non-lethal) also makes them ideal candidate biomarkers of accumulated radiation exposure. We argue that molecular analysis of breakpoint junctions formed as the result of translocations and inversions is vital to understanding the process of exchange aberration formation, since it is here where underlying repair/misrepair pathways leave their “molecular fingerprints.” Regarding relevance to NASA’s concerns, the study of chromosome aberrations stands to tell us much about mechanisms underlying the cancer process itself. The relationship between particle energy/track structure and radiogenic changes to the genome represents an important first step in understanding 1) basic dose-response relationships at low fluences and 2) fundamental carcinogenic processes that may ultimately form the basis for subsequent mitigation strategies.
Task Progress:	We continued to sequence the chromosome exchange junctions of clones whose initial progenitor cell we either exposed to low (linear energy transfer/LET) 662 keV/um gamma or to High LET 1 GeV accelerated 56-Fe ions. After sequencing the breakpoint junctions of some three dozen clones, we conclude the following: 1. In terms of the number of nucleotides excised, the level of base pair homology, and breakpoint location with respect to repetitive DNA regions, we find no significant differences in the damage resulting from these two radiation types. This is despite their LETs differing by orders of magnitude. Our results suggest that the higher relative biological effectiveness (RBE) associated with high LET radiations are not due to differences in the chemical structure within initiating DNA double-strand breaks (DSBs). Instead, we surmise that RBE/LET relationships derive from the optimal spacing between individual DSBs in the context of genomic distances.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Bailey SM, Kunkel SR, Bedford JS, Cornforth MN. "The central role of cytogenetics in radiation biology." Radiat Res. 2024 Aug 1;202(2):227-59. https://doi.org/10.1667/RADE-24-00038.1 ; PMID: 38981612 , Aug-2024